

Vision Transformer (ViT)

Deep Learning

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Contents of This Video

In this video, we will cover:

- Applying Transformers to images
- Image patches as tokens
- The ViT architecture
- The [CLS] token for classification
- When ViT beats CNNs

The Key Insight: Image Patches as Tokens

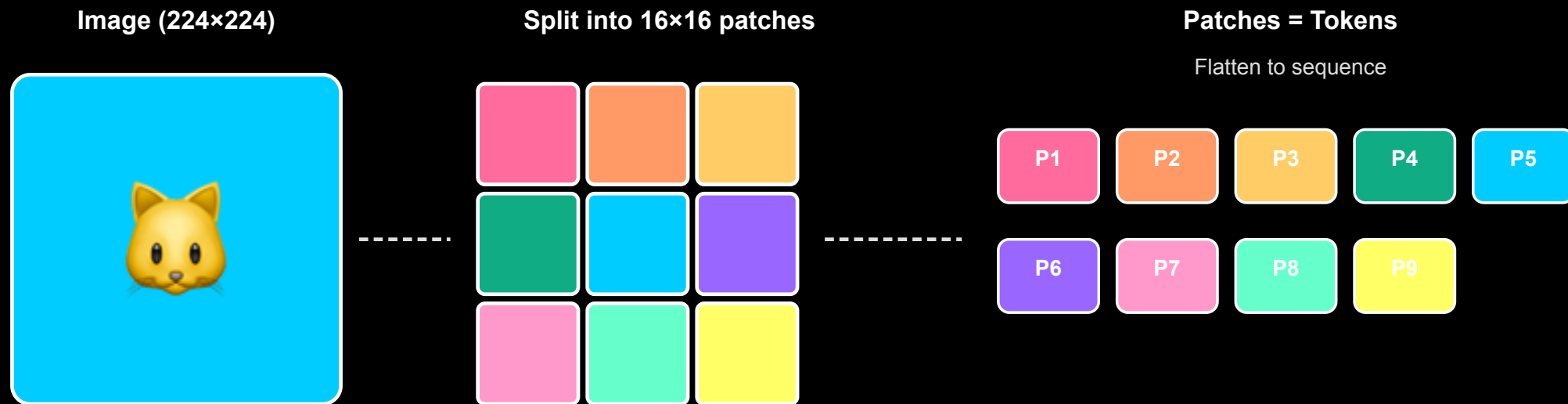
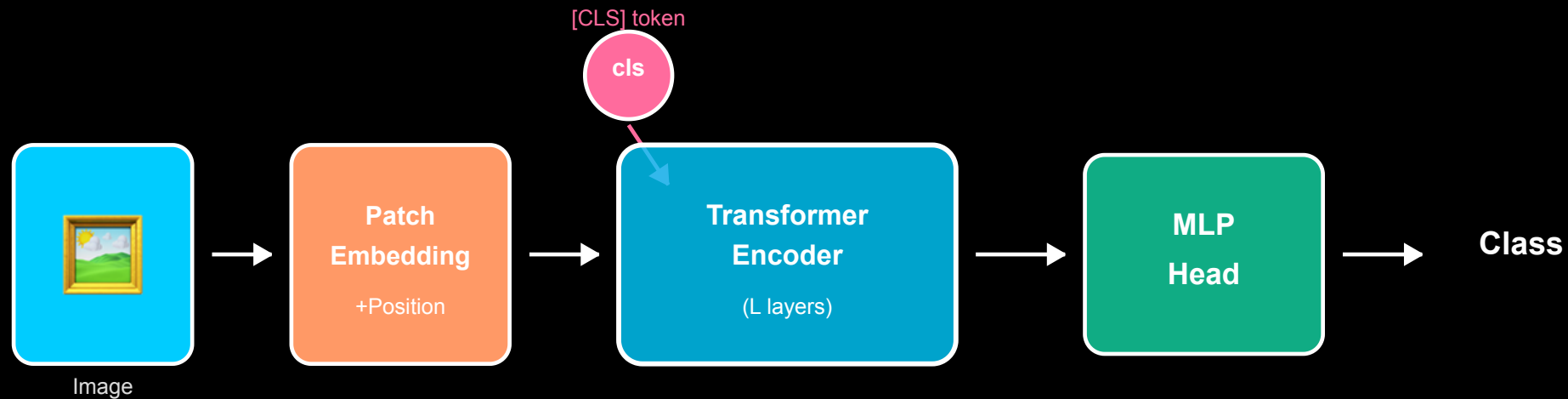


Image patches are the “words” for vision!

ViT Architecture

Vision Transformer (ViT) Architecture



Patch Embedding

Each patch $\rightarrow$ vector:

1. Flatten $16 \times 16 \times 3$ patch to 768 values
2. Linear projection to embedding dimension D
3. Add learnable position embedding

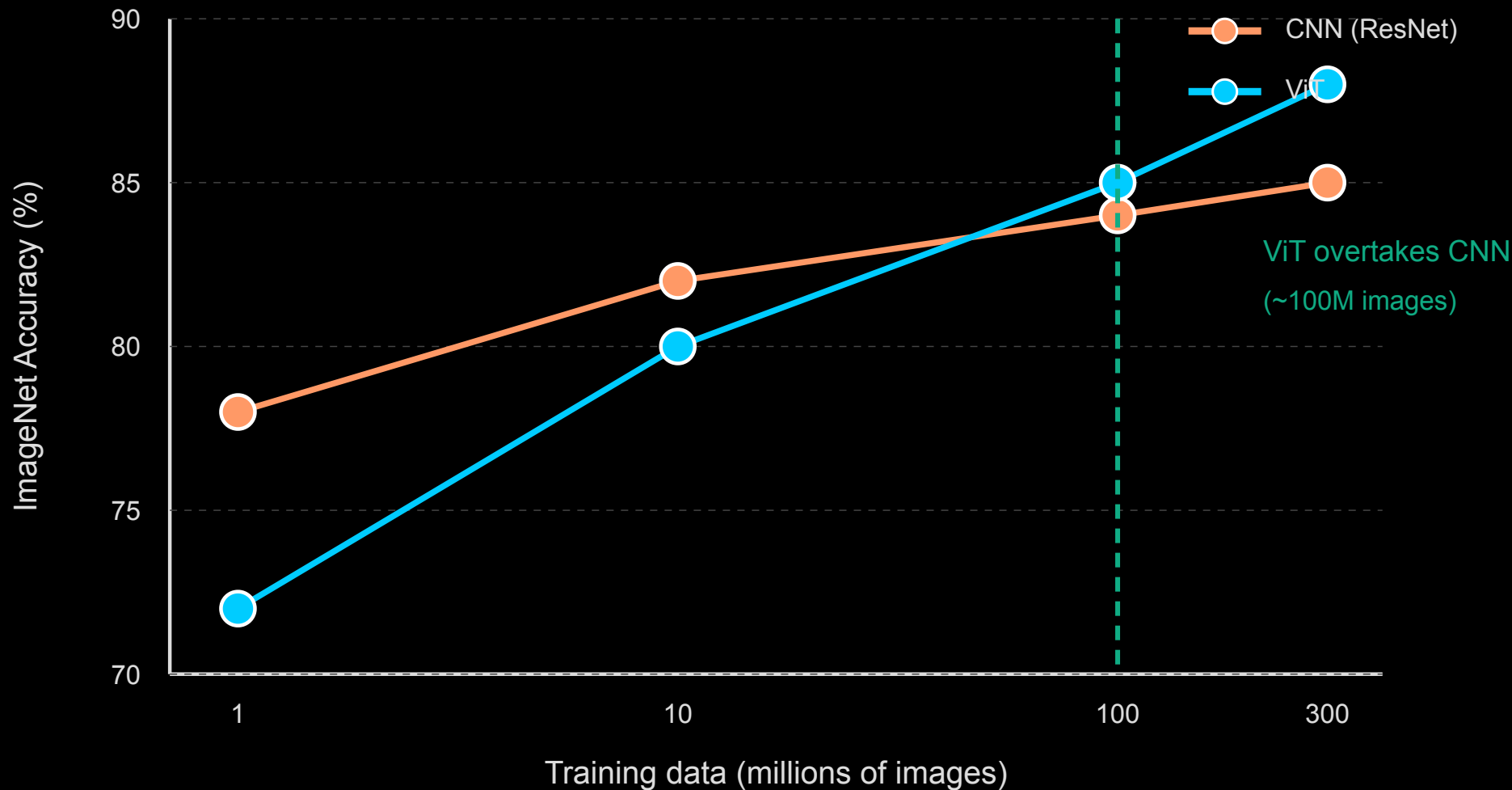
$$\mathbf{z}_0 = [x_{cls}; x_1 E; x_2 E; \dots; x_N E] + E_{pos}$$

Where:

- x_i : flattened patch i
- E : linear projection (learned)
- E_{pos} : position embeddings (learned)

ViT vs. CNN: When Does ViT Win?

ViT Needs More Data, But Scales Better



Why ViT Works

Advantages of attention over convolution:

- **Global context:** Every patch can attend to every other patch
- **Flexible:** Learns what relationships matter (not fixed by kernel)
- **Scales:** Bigger models keep improving

What ViT learns:

- Early layers: local patterns (similar to conv)
- Later layers: global relationships
- Attention patterns often interpretable

What We've Covered

In this video, we've learned:

- ViT: apply Transformers to images by treating patches as tokens
- Split image into 16×16 patches, flatten and project
- Add [CLS] token and position embeddings
- Standard Transformer encoder processes the sequence
- ViT needs more data than CNNs but scales better
- Dominant architecture for vision at large scale